THOMAS MARIANO, E.I.T. | Secret Clearance

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SKILLS

Languages: Python | SQL | MATLAB | C++ | Java | HTML | CSS | JavaScript
Systems: Linux | Windows Server | Kubernetes | Networking | Rack Design + Visio (Configuration Management)
RF & Test: SDR (ADALM-Pluto) | GNU Radio | RF bring-up & troubleshooting (SDR/analog chain) | Spectrum analysis
Design/Tools: SolidWorks | Fusion 360 | 3D Printing (Slicers) | DFAM | Requirements & procedures documentation

WORK EXPERIENCE

Systems Engineer, Senior Associate, *NASA* - *GSFC - Peraton (contractor)* **October 2024 - Present**

Supporting Human Space Flight Missions as a Search and Rescue Engineer

- Designed and developed a Python-based mission scheduling application that calculates mutual satellite visibility and automates tracking schedules, automatically sending them through, reducing planning from 2 days to 2 minutes
- Operate and sustain mission-critical search and rescue ground systems, including antenna electromechanical systems, computer hardware, RF signal paths, controller software, and network equipment supporting human space flight missions
- Lead search and rescue data analysis by developing tools to process large mission datasets, identify system and coverage performance trends that reduce errors during human spaceflight coverage by 99%
- Diagnose and remediate failures (hardware/software/network/RF/mechanical) using telemetry/log review and RF verification (spectrum/data analyzers); document root cause, corrective actions, and validation results
- Perform system administration for Linux/Windows server and rack infrastructure, including patching, configuration changes, controlled hardware, and connectivity/link troubleshooting for 180+ active systems
- Led data-center modernization, consolidating 10 server racks to 6, retiring outdated hardware and installing new hardware
- Built a Python-based mission readiness checklist adopted for human space flight missions to standardize pre-event validation, capture coverage status, and reduce operator error by 27% through repeatable pass/fail checks
- Integrating a 24-signal phased-array antenna: requirements/constraints definition, installation oversight, safety review support, and FAT planning/witnessing (procedure/checklist + discrepancy tracking) and mission support integration
- Implemented/maintained an automated monitoring console aggregating 180+ system statuses, schedules, coverage views, and camera feeds to improve network health knowledge, observability, and speed troubleshooting
- Authored original procedures and reporting templates (LOPs, EMs, weekly status/configuration reports) and maintained an issue tracker documenting anomalies and resolutions to closure, previously only 58% of coverage to now 100%
- Designed a remote-command training recovery beacon with 99% effective coverage used for human spaceflight recovery

Systems Engineer, Associate, *NASA* - *NRL - Space Systems Research Corp (contractor)* **June 2024 - October 2024**

- Designed a rocket vacuum chamber to test NASA and DOD instruments at space-like conditions

TECHNICAL PROJECTS (Full list - https://thomasmariano04.wixsite.com/portfolio)

Search and Rescue Mission Scheduler | github.com/TommyMariano/MEOSAR-Scheduler

- Developed a mission scheduling application to automate satellite coverage planning for human spaceflight operations, reducing a manual planning process from 2 days to 2 minutes
- Built an orbital mutual-visibility engine using Python to calculate satellite visibility between mission locations and ground stations using satellite data, elevation, azimuth, range, and time–based visibility constraints
- Created a front-end GUI to reduce user complication by implementing a seamless layout and creating visual graphs
- Designed numerous dynamic JSON libraries to allow modifications and modernization for the application
- Implemented a “user-friendly” scheduler editor to reduce complication with custom schedule design
- Automated a schedule export to send the proper formatting schedule directly to the scheduler, minimizing human error
- Wrote documentation and a user manual complying with Human Space Flight support

Remote Command Tri-Beacon | Human Spaceflight Recovery

- Integrated Raspberry Pi, u-blox GPS, Iridium modem, ADALM-Pluto SDR, GPIO relay, and RF amplifier into a remotely controlled training recovery beacon prototype (Python + GNU Radio)
- Designed Python and GRC control scripts and SDR configuration to improve reliability, including UTC standardization, structured logging, and elimination of tight-loop CPU saturation (100%/70% -> 0.1%/0.1% CPU usage) from older designs
- Generated and communicated custom beacon hex IDs and SDR parameters to meet protocol requirements
- Implemented/validated Iridium remote command functions: start/stop transmit, start logging/recording, and time-check

EDUCATION

University of Maryland, Clark School of Engineering - B.S. in Mechanical Engineering, College Park, MD **May 2024**

Anne Arundel Community College - Computer Science **Current**

CERTIFICATIONS

FE Mechanical (E.I.T.) – Passed **May 2025**

Georgia Tech (edX) Verified Certificates (3): Computing in Python; OOP in Java; Data Structures & Algorithms **April 2026**

AWARDS

Robert H. Goddard Team Award - NASA, Supporting the Artemis II mission **June 2026**